1) Introduction to Stochastic Gene Expression

Gene expression involves various interactions

- Transcription factors binding to promoter

- RNA polymerase transcribing mRNA

- Translation of mRNA into proteins

The models we have used so far involve deterministic dynamics, where the interactions denoted above happen in exactly the way they are described.

These models tend to predict average behavior (mean field theorem),

however in small-scale systems (aka; gene expression!), randomness can have a significant effect

Stochastic gene expression refers to the fact that these processes happen randomly at different times.

Why Do We Need Stochastic Models?

If the number of molecules involved is large, deterministic models work well eg; small random fluctuations in the number of bacteria in an ocean model

However, when molecule numbers are small - random fluctuations can drastically affect observed dynamics!

A single gene turning ON or OFF can have large consequential effects eg; large variations in protein numbers between cells

The main computational take away from the next 2 weeks of labs will be the Gillespie Algorithm.

In short, the Gillespie Algorithm:

1) Determines when the next event will occur

2) Chooses which event (e.g., production or degradation) will happen

3) Updates the system state and repeats the process.

BUT before diving into the algorithm itself, we first need to understand a key mathematical concept:

poisson processes and waiting times --> main part of todays lecture

2) Poisson Processes and Waiting Time Theory

A Poisson process models events that occur randomly over time, such as mutations or molecule production/degradation

In a Poisson process, events occur randomly over time with a fixed average rate.

However, the exact time of the next event is unknown—it follows an exponential distribution

This means that while we can calculate the average waiting time, the actual time fluctuates

Example:

Let's say an event happens at a rate b (e.g., 5 times per hour)

We want to know the probability that the next event occurs exactly at time t

Logically this means:

1) It should not happen before time t—meaning, in every tiny time step dt, the event does not occur

2) It must occur at time t—meaning, in the tiny time step from t to t+dt, the event does occur

Mathematically this means:

- The probability that the event does not happen in a small time step dt is (1−bdt).

- If we divide time into many small intervals from 0 to t, the event must not occur in any of those intervals.

- The probability of this happening is approximately e^{-bt} which decreases over time—this is why long waiting times are less likely.

- Finally, multiplying this by the probability that the event does occur at t, we get:

$$p(t) = \overbrace{e^{-bt}}^{\text{prob of event not happening till t}} * \overbrace{b}^{\text{prob of event happening at t}}$$

$$p(t) = be^{-bt}$$

This is the exponential distribution, which tells us that short waiting times are more likely than long ones.

This is the KEY result: A process that occurs at a rate b leads to exponentially distributed events

This is termed a Poisson process --> one in which the average time between events is fixed,

but the exact time of the next event is unknown and sampled from a memory-less (exponential) distribution.

Now it is time to generate exponentially distributed random numbers

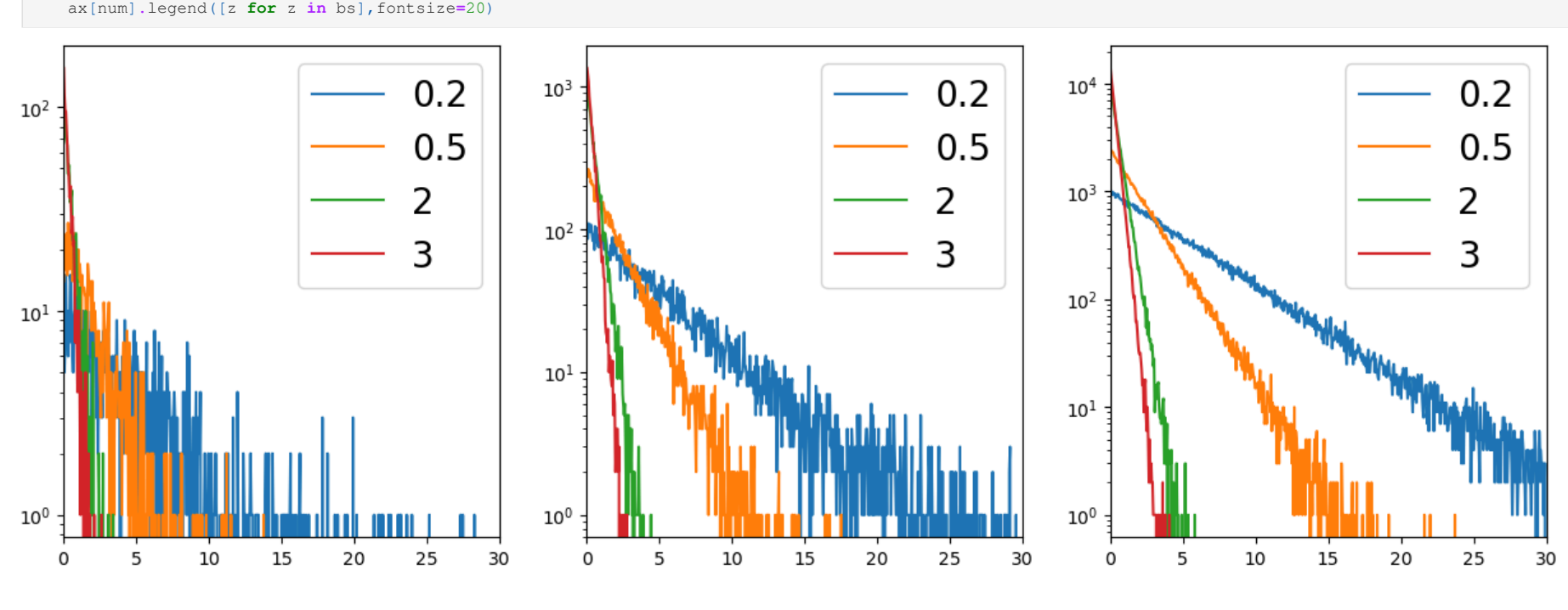
In [6]:
```
import numpy as np
import matplotlib.pyplot as plt
b=3
np.random.exponential(1/b)
#Here, the rate is 3 per hr, and the function np.random.exponential takes a single input - the inverse of the rate
# and returns an exponentially distributed random number.
```

Out[6]:
```
0.012629366692069885
```

Challenge Problem: Exponentially distributed random numbers

Plot the probability distribution for 104 and 105 exponentially sampled random numbers. Are the realized distributions in fact exponential? Now, change the rate parameter, e.g., setting it to b = 0.2, 0.5, 2 or 3. Recall that the hist command is one way to visualize distributions, although there are other options as well. If your code works, the result should look something like the following:

In [8]:
```
import numpy as np
import matplotlib.pyplot as plt
row,col=(1,3)
fig,ax=plt.subplots(row,col,figsize=(col*5,row*5))
nums=[10**3,10**4,10**5]
bs=[0.2,0.5,2,3]
bi=np.arange(0,30.05,0.05)

for num in range(len(nums)):
    n=nums[num]
    for b in bs:
        d=np.random.exponential(1/b,n)
        ns, bin_edges=np.histogram(d,bins=bi)
        bin_mids=(bin_edges[1:] + bin_edges[:-1])/2
        ax[num].plot(bin_mids,ns)
    ax[num].set_yscale('log')
    ax[num].set_xlim(0,30)
    ax[num].legend([z for z in bs],fontsize=20)
```



The code generates random samples of sizes $10^3, 10^4, \text{and } 10^5$ to analyze their probability distribution functions (PDFs).

It visualizes the distributions using histograms with logarithmically scaled y-axes, revealing their exponential nature.

The slopes of the plotted lines represent the rate parameter (b)

which is close to but not exactly the generating value.

However, the 95% confidence intervals (CIs) are expected to include the true mean.

Understanding Bootstrapping for Estimating Confidence Intervals

In simulations or experiments, we often want to estimate parameters (e.g., the average time between events or the rate of a process) and quantify the uncertainty in our estimates.

One way to do this is by bootstrapping, a statistical method that uses random resampling to approximate the confidence interval (CI) of an estimate.

Now work through the bootstrapping/confidence interval part on your own - not much to teach here!

In [11]:
```
num=10**5
bvals=[.2,.5,2,3]
b_est=np.zeros(len(bvals))
b_ci=np.zeros((len(bvals),2))
#loop for b
for i in range(len(bvals)):
    b=bvals[i]
    x=np.random.exponential(1/b,num)
    #mean
    b_est[i]=1/np.mean(x)
    tmpn=1000
    bootstrap_est=np.zeros(tmpn)
    for j in range(tmpn):
        #sample from x, allow repeats
        tmpind=np.random.randint(num,size=num)
        bootstrap_x=x[tmpind]
        bootstrap_est[j]=1/np.mean(bootstrap_x)
    bootstraps_sorted=np.sort(bootstrap_est)
    b_ci[i,0]=bootstraps_sorted[int(tmpn*0.025)]
    b_ci[i,1]=bootstraps_sorted[int(tmpn*0.975)]
b_ci
```

Out[11]:
```
array([[0.19900408, 0.20137454],
       [0.49650684, 0.50233221],
       [1.99001446, 2.01365304],
       [2.97723954, 3.01478291]])
```

3) Multiple Stochastic Events

Now we know how to calculate our waiting times for the next Poisson event to happen

But.. what if there are multiple events in the same system? How do we calculate our waiting time to the next event

And also.. which event happens first?

Specifically, what if we have two competing events:

- Protein production at rate b_1

- Protein degradation at rate b_2

In [13]:
```
# Set rates
b1 = 0.5   # Protein production rate
b2 = 2.5   # Protein degradation rate

# Generate random waiting times for each event
r1 = np.random.exponential(1/b1)
r2 = np.random.exponential(1/b2)

# The event that happens first
next_event_time = min(r1, r2)

print(f"The next event occurs at time {next_event_time:.3f} hours")
```

```
The next event occurs at time 1.207 hours
```

Intuitively try doing this part on your own,

understand why you will mostly get your next event happen closer to a certain value (use a for loop as suggested in the text)

In [15]:
```
# Generate a random number between 0 and 1

for i in range(15):
    # Determine which event occurs
    U = np.random.rand()
    if U < (b1 / (b1 + b2)):
        event = "Protein Production"
    else:
        event = "Protein Degradation"

    print(f"The first event is: {event}")
```

```
The first event is: Protein Degradation
The first event is: Protein Degradation
The first event is: Protein Degradation
The first event is: Protein Degradation
The first event is: Protein Production
The first event is: Protein Degradation
The first event is: Protein Degradation
The first event is: Protein Degradation
The first event is: Protein Degradation
The first event is: Protein Degradation
The first event is: Protein Degradation
The first event is: Protein Production
The first event is: Protein Degradation
The first event is: Protein Degradation
The first event is: Protein Degradation
```

Now that we understand how multiple stochastic events are timed,

we can compare our simulated results with the theoretical expectation.

We can once again say that the waiting time until the first event happens follows an exponential distribution $p(t) = e^{-bt}$

But importantly this time with a rate $b = b_1 + b_2$

Where the probability that event 1 happens first is $\frac{b_1}{b_1+b_2}$ and the probability that event 2 happens first is $\frac{b_2}{b_1+b_2}$

We should see this pattern in our simulation.

A useful way to analyze and compare distributions is through the Cumulative Distribution Function (CDF).

Cumulative Distribution Function (CDF) gives the probability that an event occurs by a certain time t,

rather than at an exact time (aka at or before a certain time t)

Mathematically, the CDF for an exponential distribution with rate is

$$Q(t) = 1 - e^{-bt}$$

This function tells us how likely it is that the event has already happened by time t

Eg; If Q(1)=0.7, this means there is a 70% chance that an event has already occurred before t=1

As t increases, Q(t) approaches 1 (event certainty)

How Do We Compare the CDF of Simulated Data?

1) Run many simulations of the event process and record the waiting times.

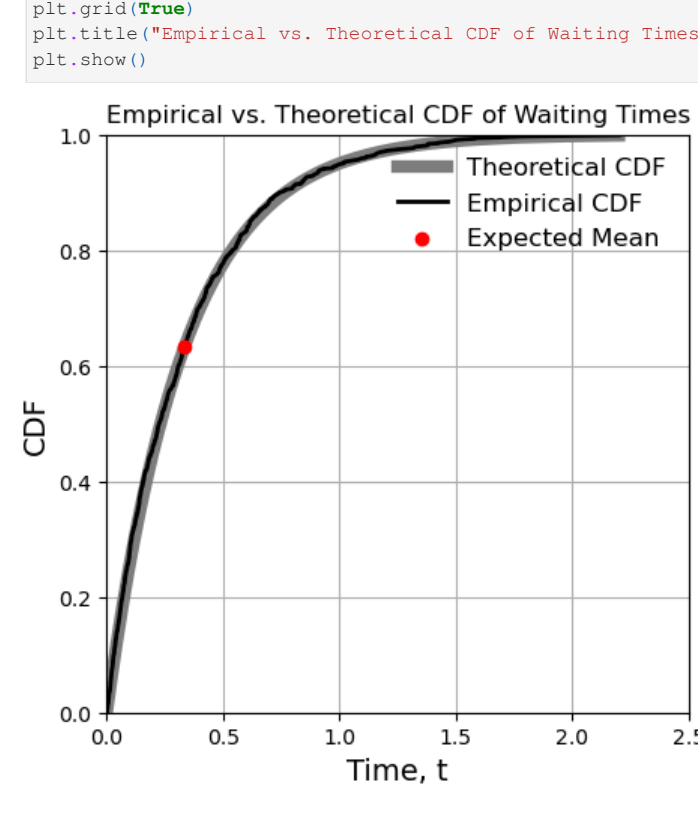
2) Sort the waiting times and calculate how many events occurred before each time (this gives the empirical CDF).

3) Plot both the empirical CDF and the theoretical CDF for comparison.

We will now generate 1000 simulations, extract the waiting times,

and compare the empirical distribution to the theoretical expectation.

In [20]:
```
# Define event rates
b1 = 0.5   # Protein production rate
b2 = 2.5   # Protein degradation rate
n = 1000   # Number of simulations

# Generate waiting times for both events
x1 = np.random.exponential(1/b1, n)
x2 = np.random.exponential(1/b2, n)

# Find the time of the first event in each simulation
etime = np.minimum(x1, x2)  # Element-wise min operation

# Sort event times to build the empirical CDF
sortdata = np.sort(etime)
cdfvals = np.arange(n) / n  # Fraction of values less than each sorted value

# Define the theoretical CDF for an exponential process with rate b1 + b2
# remember lambda is just a short form function that takes 1 parameter
poiss3cdf = lambda t: 1 - np.exp(-(b1 + b2) * t)

# Plot the empirical CDF vs. theoretical expectation
plt.figure(figsize=(5,5))
plt.plot(sortdata, poiss3cdf(sortdata), color='gray', linewidth=6, label="Theoretical CDF")
plt.plot(sortdata, cdfvals, color='k', linewidth=2, label="Empirical CDF")
plt.scatter(1/(b1+b2), poiss3cdf(1/(b1+b2)), color='red', label="Expected Mean", zorder=3)

plt.xlabel("Time, t", fontsize=14)
plt.ylabel("CDF", fontsize=14)
plt.legend(fontsize=12, frameon=False, loc="upper right")
plt.xlim(0, 2.5)
plt.ylim(0, 1)
plt.grid(True)
plt.title("Empirical vs. Theoretical CDF of Waiting Times")
plt.show()
```



The gray curve represents the theoretical CDF based on $b = b_1 + b_2 = 3$ per hour

The black curve represents the empirical CDF from 1000 simulated waiting times given the two events and their rates

The red dot marks the expected mean waiting time $1/(b_1 + b_2)$, showing where the average event should occur.

Conclusions:

- The empirical CDF closely follows the theoretical CDF, confirming that our simulation follows the expected Poisson process.

- The average event time is shorter than for either process alone, since multiple competing events increase the likelihood of something happening sooner.

- This confirms that rates of events add together, and the timing of the first event follows an exponential distribution with rate

Challenge Problem: Identity of Random Events

This challenge is centered on solving the following: given two independent Poisson processes operating concurrently, what fraction of times do each of the two processes take place first? That is to say, if b1 = 0.5 and b2 = 2.5, what fraction of events are associated with process 1 and what fraction of events are associated with process 2? Write your own code to try and solve this, leveraging the code already written thus far in the laboratory guide.

In [24]:
```
# Generate random numbers
num=10**3
b1=0.5
b2=2.5
b1_rnd=np.random.exponential(1/b1,num)
b2_rnd=np.random.exponential(1/b2,num)
# Find out which came first
tmpi1=np.argwhere(b1_rnd<b2_rnd)
tmpi2=np.argwhere(b2_rnd<=b1_rnd)
print('Fraction of 1-events that come first is',len(tmpi1)/num)
print('Fraction of 2-events that come first is',len(tmpi2)/num)
```

```
Fraction of 1-events that come first is 0.161
Fraction of 2-events that come first is 0.839
```

Takeaways: if $b_1 = 0.5$ and $b_2 = 2.5$, event 1 occurs first $b_1/(b_1 + b_2)$ = 1/6th of the time and event 2 occurs 5/6th of the time

The simulation confirms the theoretical expectation—the fraction of times each event occurs first matches their relative rates

Key Takeaways

- Instead of always picking the shortest waiting time,

- we can first determine when any event will occur using a Poisson process with a total rate equal to the sum of all individual event rates.

- After determining the time of the next event, we select which specific event happens based on their relative probabilities.

- Events with higher rates are more likely to occur first.

- This method is the core principle of the Gillespie Algorithm, allowing efficient and accurate simulation of stochastic biochemical and cellular processes

In [28]:
```
#### Lesson Split
```

Introduction to the Gillespie Algorithm in Gene Expression

The Gillespie Algorithm is a widely used method for simulating stochastic processes in biology and chemistry.

In gene expression, molecules like mRNA and proteins are produced and degraded at random times, making deterministic models inaccurate for small molecule numbers.

Instead of assuming continuous changes, the Gillespie Algorithm models individual events step by step, considering:

1) When the next event occurs.

2) Which event happens (e.g., production or degradation).

3) How the system updates after the event

Our work last week gave us very pretty steady states eg; $p^* = \beta/\alpha$

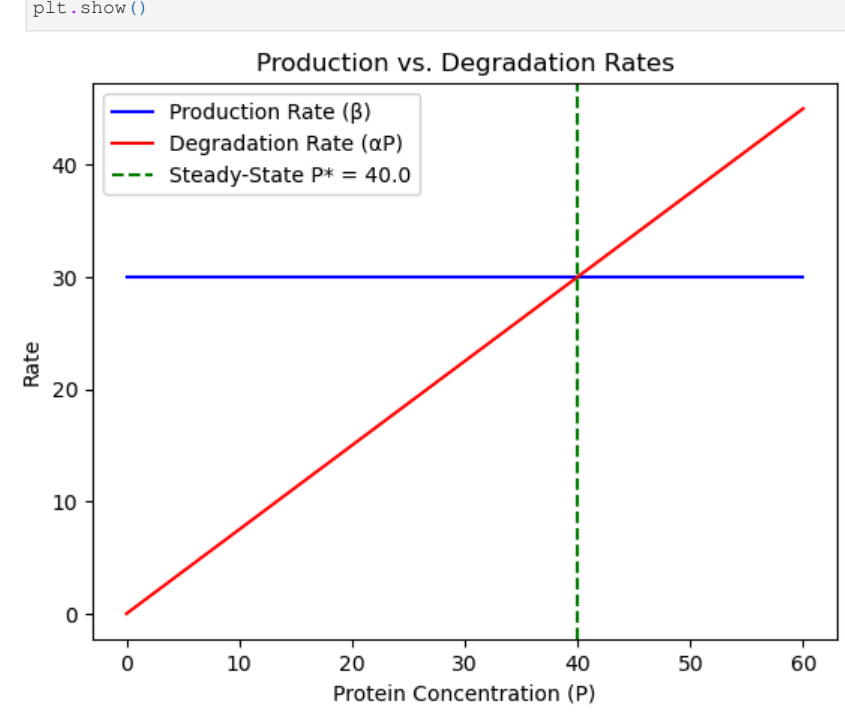
However how realistic is this? Can we really have "decimal place" protein levels? What is there is random fluctuation in production?

According to the Gillespie algorithm, the total rate at which any event occurs is the sum of the rates of individual processes:

Our individual process are production β and decay αp

Plotting these two processes as a function of p gives us an intersection, can someone tell me what this value intuitively is?

In [33]:
```
beta = 30  # Production rate
alpha = 0.75  # Degradation rate
P_vals = np.linspace(0, 60, 100)  # Protein concentration values


production_rate = np.full_like(P_vals, beta)  # Constant beta
degradation_rate = alpha * P_vals  # Linear degradation


plt.plot(P_vals, production_rate, label="Production Rate (β)", linestyle="-", color="blue")
plt.plot(P_vals, degradation_rate, label="Degradation Rate (αP)", linestyle="-", color="red")


P_star = beta / alpha
plt.axvline(P_star, color="green", linestyle="dashed", label=f"Steady-State P* = {P_star}")


plt.xlabel("Protein Concentration (P)")
plt.ylabel("Rate")
plt.title("Production vs. Degradation Rates")
plt.legend()
plt.show()
```



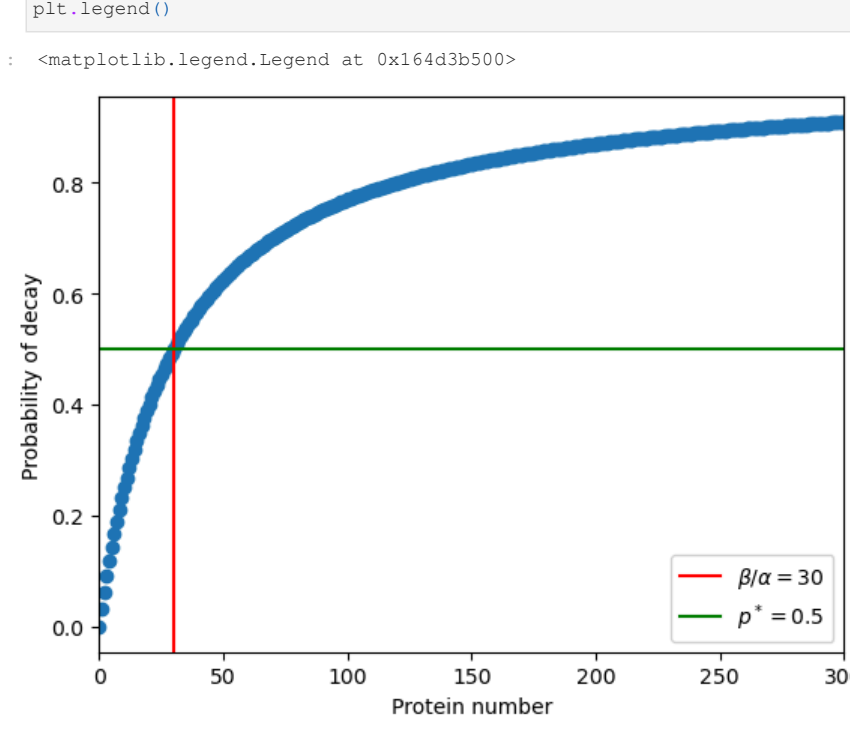
Challenge Problem: State-Dependent Rates

Write code that calculates the probability of decay as a function of the state p. Using this code, calculate the probability of decay at equilibrium, i.e., when $p^* = \beta/\alpha = 30/1$.

Use code and visualizations to support your argument. If your code is working, it should look like this, depending on whether you visualize from 0 to 60 proteins (left) or from 0 to 300 (right):

In [36]:
```
alpha = 1
r = 30
probdecay = lambda p: alpha*p/(r+alpha*p)
# Plot the probability of decay as a function of proteins
pvec = np.linspace(0,300,300)
plt.scatter(pvec,probdecay(pvec))
plt.xlim(0,300)
plt.xlabel('Protein number')
plt.ylabel('Probability of decay')
plt.axvline(30, label = r'$\beta / \alpha = 30$', color = 'red')
plt.axhline(0.5, label = r'$p^* = 0.5$', color = 'green')
plt.legend()
```

Out[36]:
```
<matplotlib.legend.Legend at 0x164d3b500>
```



The probability of a decay event taking place before a production event is the ratio of the decay rate to the total rate of all processes.

Hence, it is $\alpha p/(\beta + \alpha p)$ -> which is a saturating function of p.

Note that at equilibrium then $\beta = \alpha p^*$ such that by definition, the probability of decay is 1/2, the probability of production is 1/2,

and so the system is equally likely to increase protein levels by 1 as it is to decrease protein levels by 1

How the Gillespie Algorithm Works with Gene Expression

Step-by-Step Breakdown (read alongside pseudocode and try coding up)

1) Initialize the system: set time range and set initial values of t= 0, P = 0

2) Compute event rates:

- Production rate: β

- Degradation rate: αP

3) Determine the waiting time until the next event

- Draw a random waiting time from an exponential distribution with rate β + αP (recall that we add the rates if we are waiting for next event)

4) Determine which event occurs

- Calculate the probability of production or degradation eg; P(prod) = $\frac{\beta}{\beta+\alpha P}$

- Randomly decide if production or degradation occurs

5) Apply event and update state accordingly

6) Store updated state

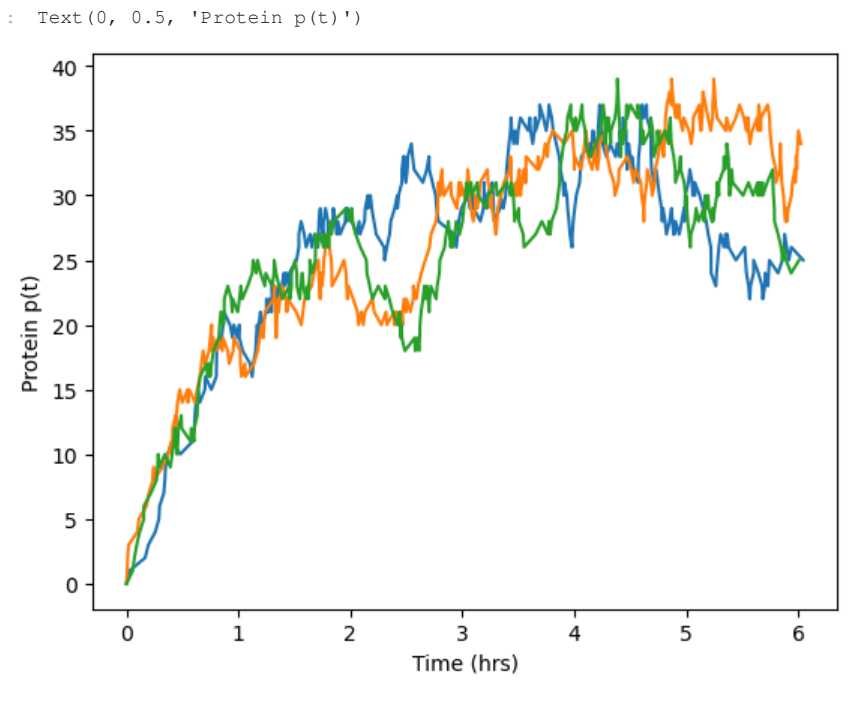
7) Repeat steps 2-6 in a loop till you reach your time limit

Now follow the pages in the booklet, and plot your gillespie algorithms to reproduce figure right above bonus

Note: save the gillespie algorithm as a FUNCTION (booklet doesn't do that)

In [68]:
```
# Define the stochastic simulation function with a time range
def stochsim_protein(trange, p0, beta, alpha):
    currp = p0  # Initial protein count
    currt = trange[0]  # Start time
    tvals = [currt]
    pvals = [currp]

    while currt < trange[1]:  # Use trange instead of tmax
        # Calculate rates
        decayrate = alpha * currp
        prodrate = beta
        total_rate = decayrate + prodrate

        # Find waiting time
        deltat = np.random.exponential(1 / total_rate)
        currt += deltat
        prob_produce = prodrate / total_rate

        # Update state
        if np.random.uniform() < prob_produce:
            currp += 1
        else:
            currp -= 1

        # Record state
        tvals.append(currt)
        pvals.append(currp)

    return np.array(tvals), np.array(pvals)

# Set parameters
tmax = 6
p0 = 0
alpha = 1
beta = 30


# Run the simulation 3 times with range-based time input
for i in range(3):
    t, p = stochsim_protein([0, tmax], p0, beta, alpha)  # Use time range [0, tmax]
    plt.plot(t, p)
plt.xlabel('Time (hrs)')
plt.ylabel('Protein p(t)')
```

Out[68]:
```
Text(0, 0.5, 'Protein p(t)')
```



Takeaways:

- This function accurately simulates gene expression using the Gillespie Algorithm

- The random nature of the process leads to different protein levels in each simulation

- While random, the steady-state protein count fluctuates around $p^* = \beta/\alpha = 30$ but individual trajectories vary due to stochastic effects.

On your own, try out the new variations they talk about - note that the differences and explore the system!

Sampling of Stochastic Trajectories

- Stochastic trajectories are hard to compare because events happen at random intervals instead of fixed time points

- Experimental data is collected at regular intervals, making direct comparison b/w experiments and stochastic trajectories challenging

- Resampling stochastic trajectories at fixed times helps align them with experimental measurements

- This ensures better consistency and comparability across different simulations

Steps to Sample a Stochastic Trajectory at Fixed Times (read alongside pseudocode and try coding up)

1) Define a vector of fixed time points for sampling.

2) Create an empty vector to store sampled protein counts.

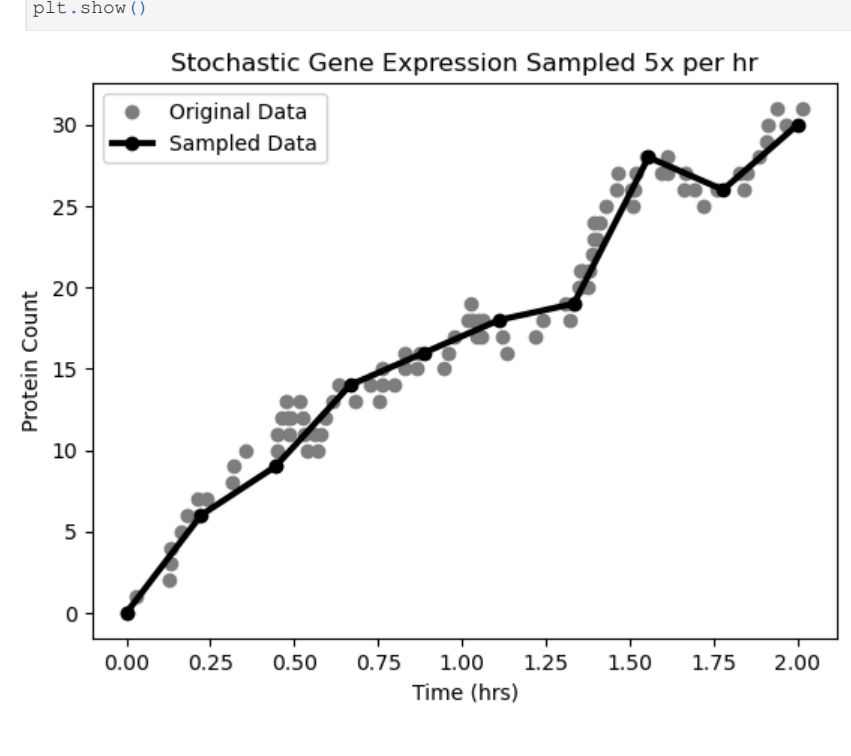
3) Loop through each fixed time point and find the most recent protein count before that time.

4) Store this value in the sampled dataset.

Challenge Problem: Discrete sampling of stochastic trajectories

Implement a function to take the output of an event-driven Gillespie algorithm and return the values of the state at discrete or otherwise pre-specified intervals. This will be useful in moving from the system perspective to the measurement perspective – see Figure 2. (right):

In [44]:
```
# Function to sample a trajectory at discrete time intervals
def sample_traj(t, y, trange):

    """ Samples a trajectory (t, y) at discrete intervals in trange """

    t = np.array(t)
    y = np.array(y)

    # Initialize arrays using np.zeros
    ts = np.zeros(np.shape(trange)) # Time points we want to sample at
    ys = np.zeros(np.shape(trange))# Protein values at sampled times

    # Initialize first sampled point
    curt = trange[0]
    ind = np.argwhere(t <= curt)
    curind = ind[-1][0]  # Extract valid index
    ts[0] = t[curind]
    ys[0] = y[curind]

#Scans across the sample interval and then moves the actual dynamics forward until we cross it
    for i in range(1, len(trange)):
        nextt = trange[i]
        while curind < len(t) and t[curind] < nextt:
            curind += 1
        ts[i] = nextt
        ys[i] = y[curind - 1]  # Take the last valid y-value

    return ts, ys

# Set parameters
alpha = 1
beta = 30
tmax = 2
p0 = 0
interval = 0.2 #sampled every interval

# Run the Gillespie algorithm (now using `stochsim`)
t, y = stochsim_protein([0, tmax], p0, beta, alpha)

# Plot the original trajectory
plt.plot(t, y, 'o', color='grey', label="Original Data")

# Sample the trajectory according to interval set
ts, ys = sample_traj(t, y, np.linspace(0, tmax, int(tmax / interval)))

# Plot the sampled trajectory
plt.plot(ts, ys, 'ko-', linewidth=3, label="Sampled Data")

# Add labels and legend
plt.xlabel("Time (hrs)")
plt.ylabel("Protein Count")
plt.title("Stochastic Gene Expression Sampled " + str(round(1/interval)) + "x per hr")
plt.legend()
plt.show()
```



What the sampling function does:

1) Takes arrays t (time) and y (protein levels) that we get from our Gillespie algorithm

2) Initialize ts (sampled time points) and ys (sampled protein levels) as zero arrays of the same size as trange.

3) Finds the first recorded protein value before the first sampling time.

4) Loop through all sampling times:

- Find the most recent protein count before or at each sampling time.

- Store the corresponding protein count.

5) Return the sampled time and protein arrays.

Steady State Analysis

Now that we have a working Gillespie algorithm and a method to sample stochastic trajectories (yay!)

the next challenge is to analyze and compare steady-state statistics in two ways:

1) Tracking a Single Long-Term Simulation:

- Run the Gillespie algorithm for 20 hours starting with 0 proteins.

- Use the final protein count as the initial condition for a new simulation.

- Run the second simulation for 100 more hours.

- Sample 1000 time points between 20 and 120 hours (i.e., every 6 minutes).

- Compute the mean and variance of the protein count over time.

- Analyze the distribution of protein numbers across time.

2) Tracking Many Independent Simulations:

- Run 1000 separate 20-hour simulations, each starting with 0 proteins.

- Record the final protein count of each simulation.

- Compute the mean, variance, and distribution of final protein numbers across these runs.

This is the crux of the next challenge problem!

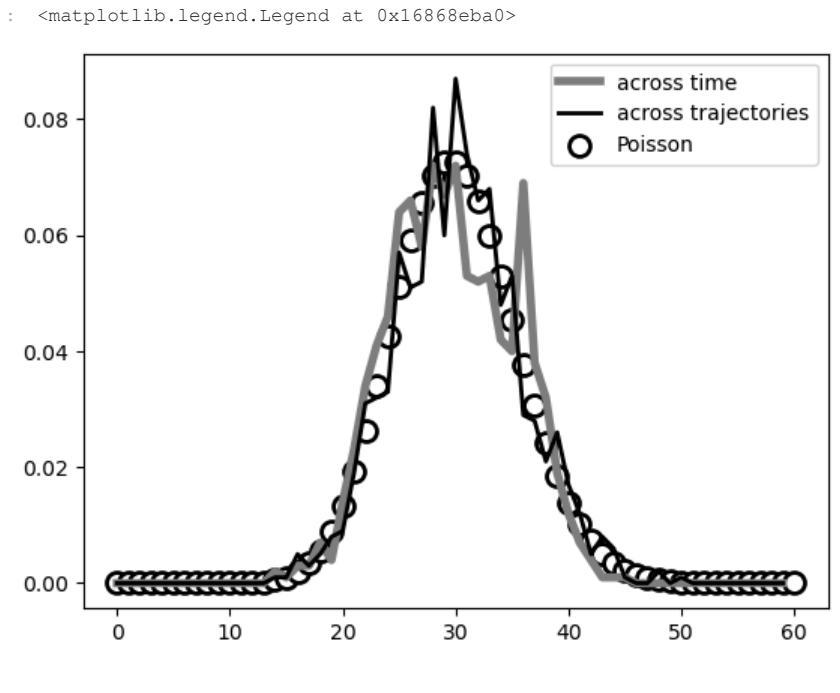
Note that challenge problem asks for poisson distribution points as well, that is a bonus (def try, but important is to show across time vs across trajectories)

Challenge Problem: Steady State Distributions of Stochastic Gene Expression

Develop code that compares and contrasts the output of the long-term sampling of a single trajectory vs. sampling of many trajectories at distinct points. Use β = 30, α = 1, and sample at the 20 hr point across 1000 runs or sample every 0.1 hrs after reaching the 20 hr point. If your code works it should look like: (below):

In [51]:
```
import scipy.stats as stats
# Parameters
beta = 30
alpha = 1

# Long term trajectory
t, y = stochsim_protein([0,20],0,beta,alpha)
y0 = y[-1]
t, y = stochsim_protein([0,100],y0,beta,alpha)
trange = np.linspace(0,100,1001)
ts, ys = sample_traj(t,y,trange)

# Histogram
py,px = np.histogram(ys,np.arange(0,61))
plt.plot(px[:-1],py/sum(py),linewidth=4,color=[0.5,0.5,0.5], label = 'across time' )

# Simulate many trajectories
numsamps = 1001
final_y0 = np.zeros(numsamps)
for j in range(numsamps):
    t, y = stochsim_protein([0,20],0,beta,alpha)
    final_y0[j] = y[-1]

py2,px2 = np.histogram(final_y0,np.arange(0,61))
plt.plot(px2[:-1],py2/sum(py2),linewidth=2,color='k', label = 'across trajectories' )

# Calculate the distribution in theory
px_theory = np.arange(0,61)
py_theory = stats.poisson.pmf(px_theory,beta/alpha)
plt.scatter(px_theory,py_theory,c='w',edgecolors='k',s=100,linewidth=2, label = 'Poisson' )

plt.legend()
```

Out[51]:
```
<matplotlib.legend.Legend at 0x16868eba0>
```



Bonus: Why Does This Lead to a Poisson Distribution?

A Poisson distribution describes the probability of observing a certain number of events in a fixed time, given a constant average rate.

For stochastic gene expression:

- The system reaches a steady state when production and degradation balance out on average

- At steady state, the number of proteins P fluctuates, but its probability follows the equation

$$P(P = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Where $\lambda = \frac{\beta}{\alpha}$ is the mean steady-state protein count

This means that, over time, the probability of finding a given number of proteins follows a Poisson distribution centered around $\lambda = \frac{\beta}{\alpha}$